

TE Curve Verification Pipeline Multi-Model Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative TE Curve Verification Pipeline TE curves by overlaying multiple model predictions on the same original measured curve. The plots are intended to show whether each model tracks the local harmonic oscillations rather than only the broad mean trend.

Scope

- each comparison image contains four deterministic held-out test curves;
- forward comparisons are shown on forward curves only;
- backward comparisons are shown on backward curves only;
- Wave 1 screening keeps the three strongest family-best models by Curve MAE [deg] within each direction;
- Original Curve uses the same visual weight as predictions and a dark-gray color for balanced comparison.

Metrics Summary

Forward Wave52r Stage5 Non Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage5_h08_seed_314159	wave52r_stage5_non_temporal	Fw	0.001694	0.002005	3.483

Forward Wave52r Stage9 Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage9_k01	wave52r_stage9_temporal	Fw	0.001374	0.001645	2.716

Forward Wave52r Stage12 Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage12_f01	wave52r_stage12_temporal	Fw	0.001444	0.001724	2.993

Forward Accepted Non Windowed Reference Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
accepted_periodic_mlp_harmonic_Fw	accepted_non_windowed_reference	Fw	0.001694	0.002008	3.439

Forward Accepted Time Windowed Incumbent Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
accepted_periodic_gru_sequence_Fw	accepted_time_windowed_incumbent	Fw	0.001618	0.001931	3.278

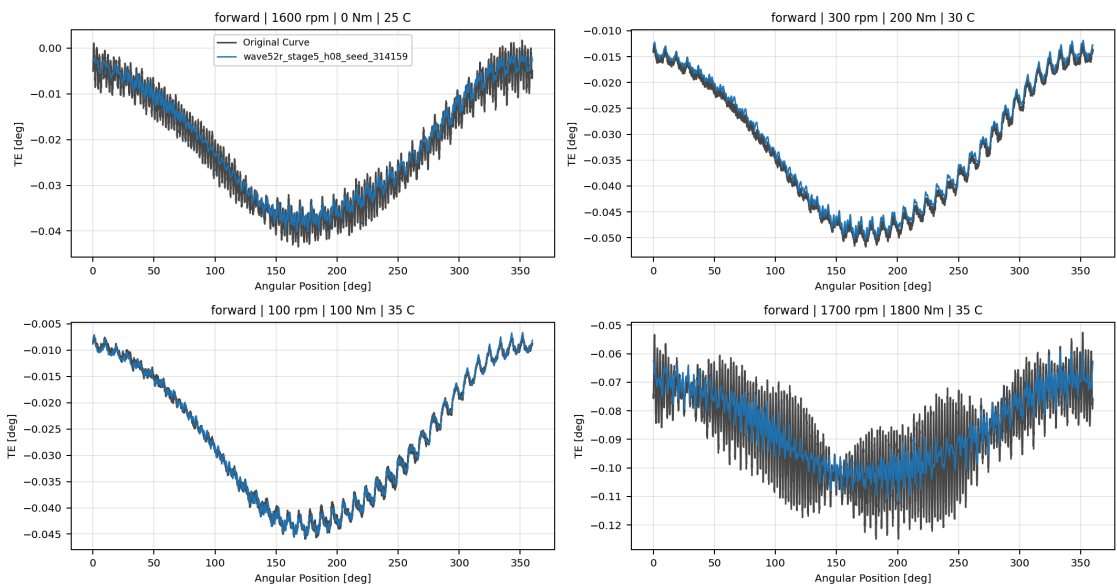
Forward Wave 5.2R Cross-Lane Finalist Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage5_h08_seed_314159	wave52r_stage5_non_temporal	Fw	0.001694	0.002005	3.483
wave52r_stage9_k01	wave52r_stage9_temporal	Fw	0.001374	0.001645	2.716
wave52r_stage12_f01	wave52r_stage12_temporal	Fw	0.001444	0.001724	2.993
accepted_periodic_mlp_harmonic_Fw	accepted_non_windowed_reference	Fw	0.001694	0.002008	3.439
accepted_periodic_gru_sequence_Fw	accepted_time_windowed_incumbent	Fw	0.001618	0.001931	3.278

Comparison Gallery - Forward Wave52r Stage5 Non Temporal Overlay

Included models: wave52r_stage5_h08_seed_314159 .

Forward Wave52r Stage5 Non Temporal Overlay

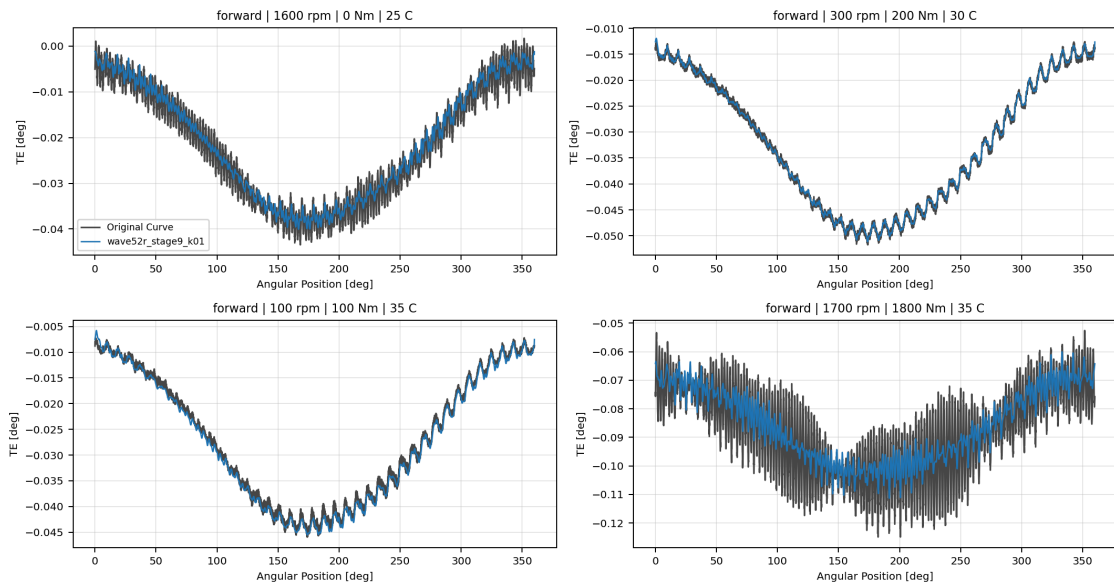


Forward Wave52r Stage5 Non Temporal Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave52r Stage9 Temporal Overlay

Included models: `wave52r_stage9_k01` .

Forward Wave52r Stage9 Temporal Overlay

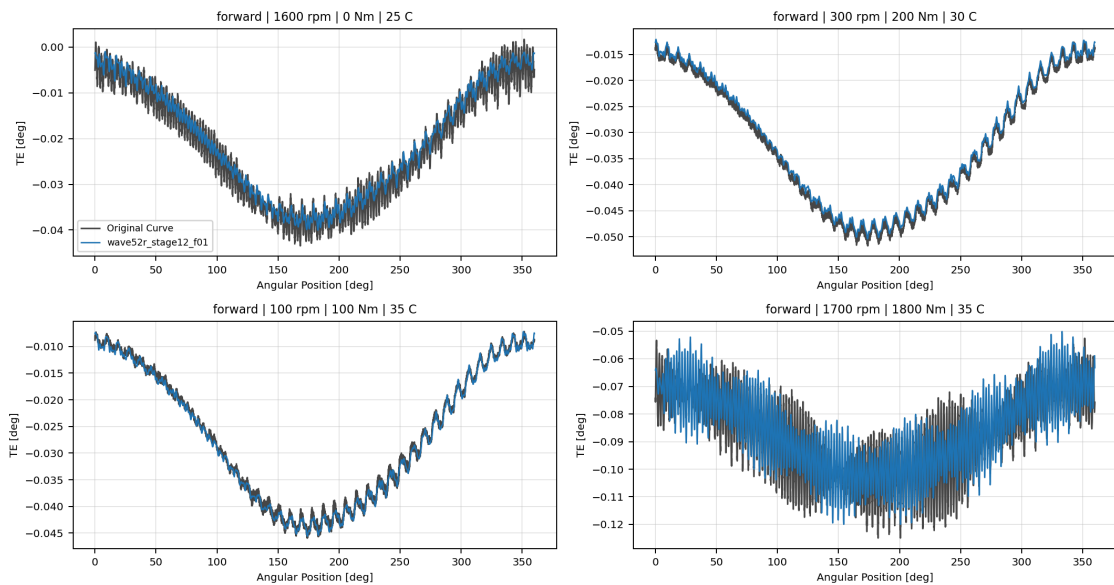


Forward Wave52r Stage9 Temporal Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave52r Stage12 Temporal Overlay

Included models: `wave52r_stage12_f01` .

Forward Wave52r Stage12 Temporal Overlay

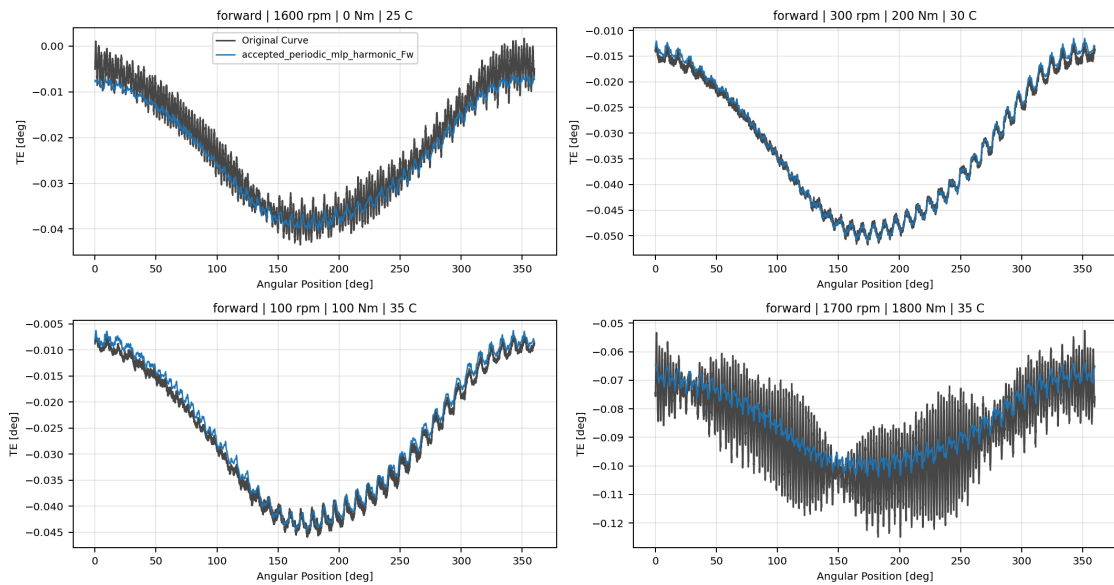


Forward Wave52r Stage12 Temporal Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Accepted Non Windowed Reference Overlay

Included models: `accepted_periodic_mlp_harmonic_Fw` .

Forward Accepted Non Windowed Reference Overlay

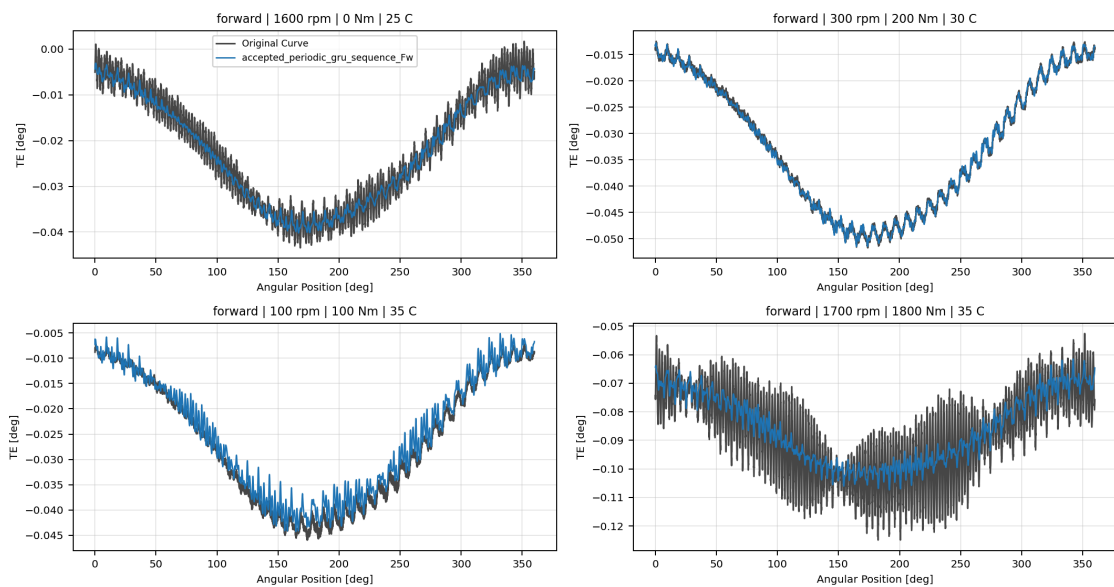


Forward Accepted Non Windowed Reference Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Accepted Time Windowed Incumbent Overlay

Included models: `accepted_periodic_gru_sequence_Fw` .

Forward Accepted Time Windowed Incumbent Overlay

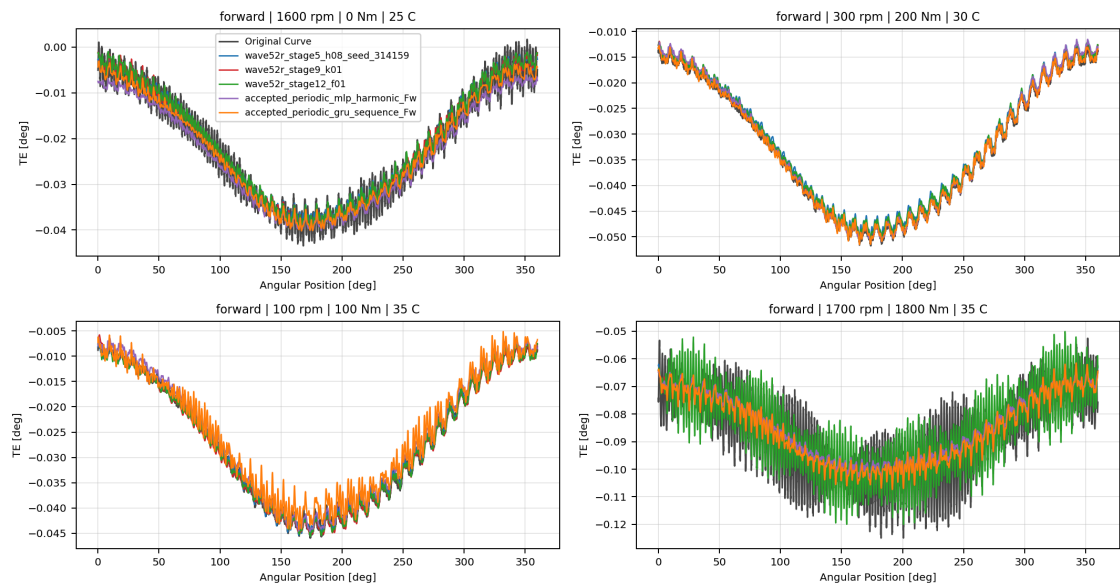


Forward Accepted Time Windowed Incumbent Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 5.2R Cross-Lane Finalist Overlay

Included models: wave52r_stage5_h08_seed_314159 , wave52r_stage9_k01 , wave52r_stage12_f01 , accepted_periodic_mlp_harmonic_Fw , accepted_periodic_gru_sequence_Fw .

Forward Wave 5.2R Cross-Lane Finalist Overlay



Forward Wave 5.2R Cross-Lane Finalist Overlay TE Curve Verification Pipeline comparison

Output Artifacts

- output directory:
`output\validation_checks\wave52r_full_candidate_track2_multi_model_curve_comparison_report\2026-07-30-11-05-54__track2_multi_model_curve_comparison_report ;`
- summary YAML:
`output\validation_checks\wave52r_full_candidate_track2_multi_model_curve_comparison_report\2026-07-30-11-05-54__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_summary.yaml ;`
- metrics CSV:
`output\validation_checks\wave52r_full_candidate_track2_multi_model_curve_comparison_report\2026-07-30-11-05-54__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_metrics.csv ;`
- report Markdown:
`doc\reports\analysis\te_curve_verification_pipeline\02_visual_reports\wave52r_full_candidate_multi_model_curve_comparison_report\[2026-07-30]\track2_multi_model_curve_comparison_report.md .`